Popularity, Prestige, and Elite Alignment in the Dating Network

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Network Science of Socio-Economic Systems

Personal Motivation

Why this project?

On Twitter/X Spaces, I kept hearing claims like:

> "The top 20% get most of the attention on dating apps."
> "The bottom 10–15% never find a partner at all."
> "Below-average men get 3–5 partners in a lifetime."

Everyone discusses this anecdotally — nobody looks at data.

I wanted to test these claims with real network data.

This project was born from that motivation: take the intuition, apply network science, and see what 17 million ratings actually say.

The rec-dating Dataset

17.4M

Total ratings in the network

135,360

Unique raters (source nodes)

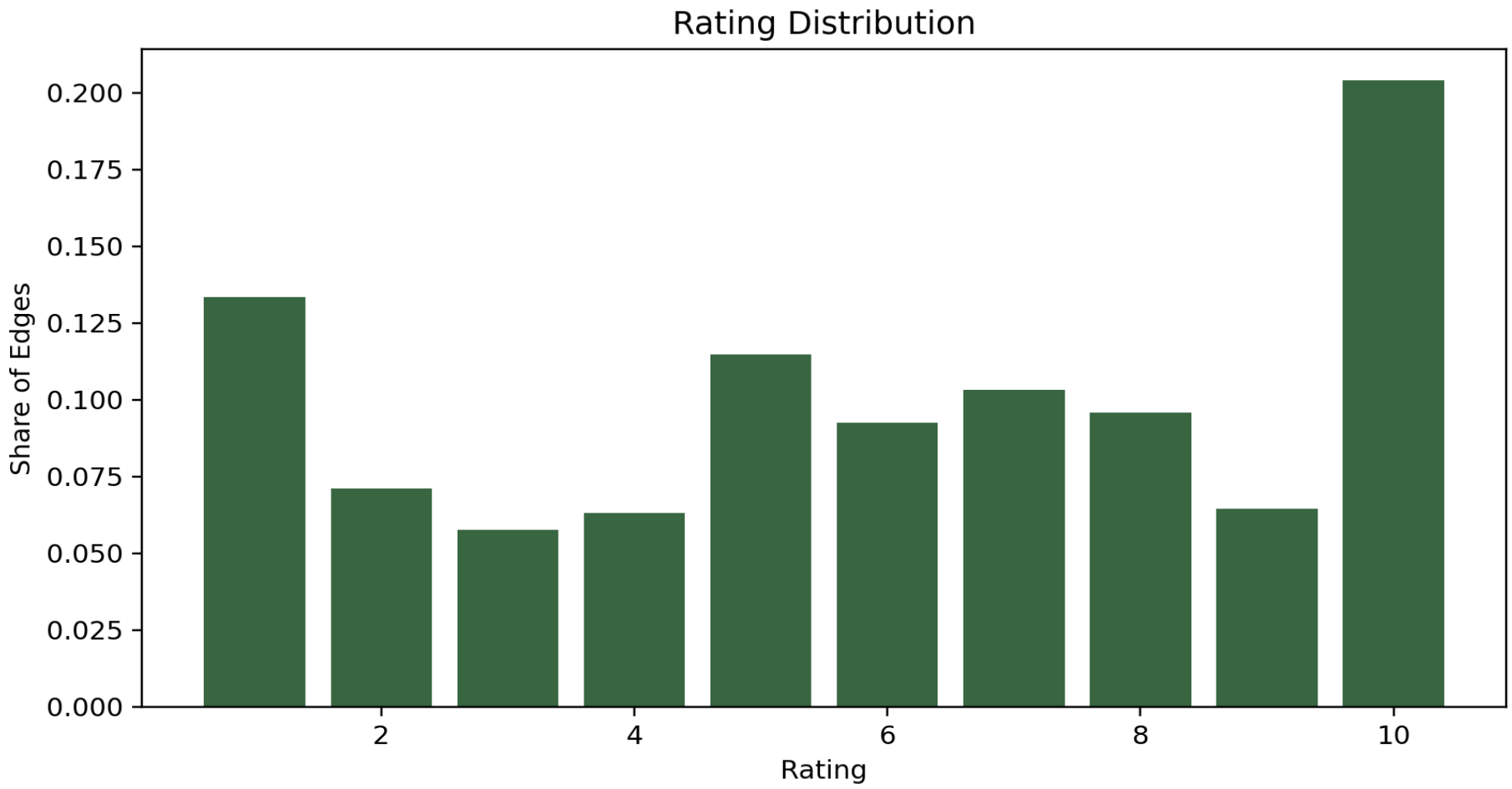
168,792

Unique profiles (target nodes)

5.94

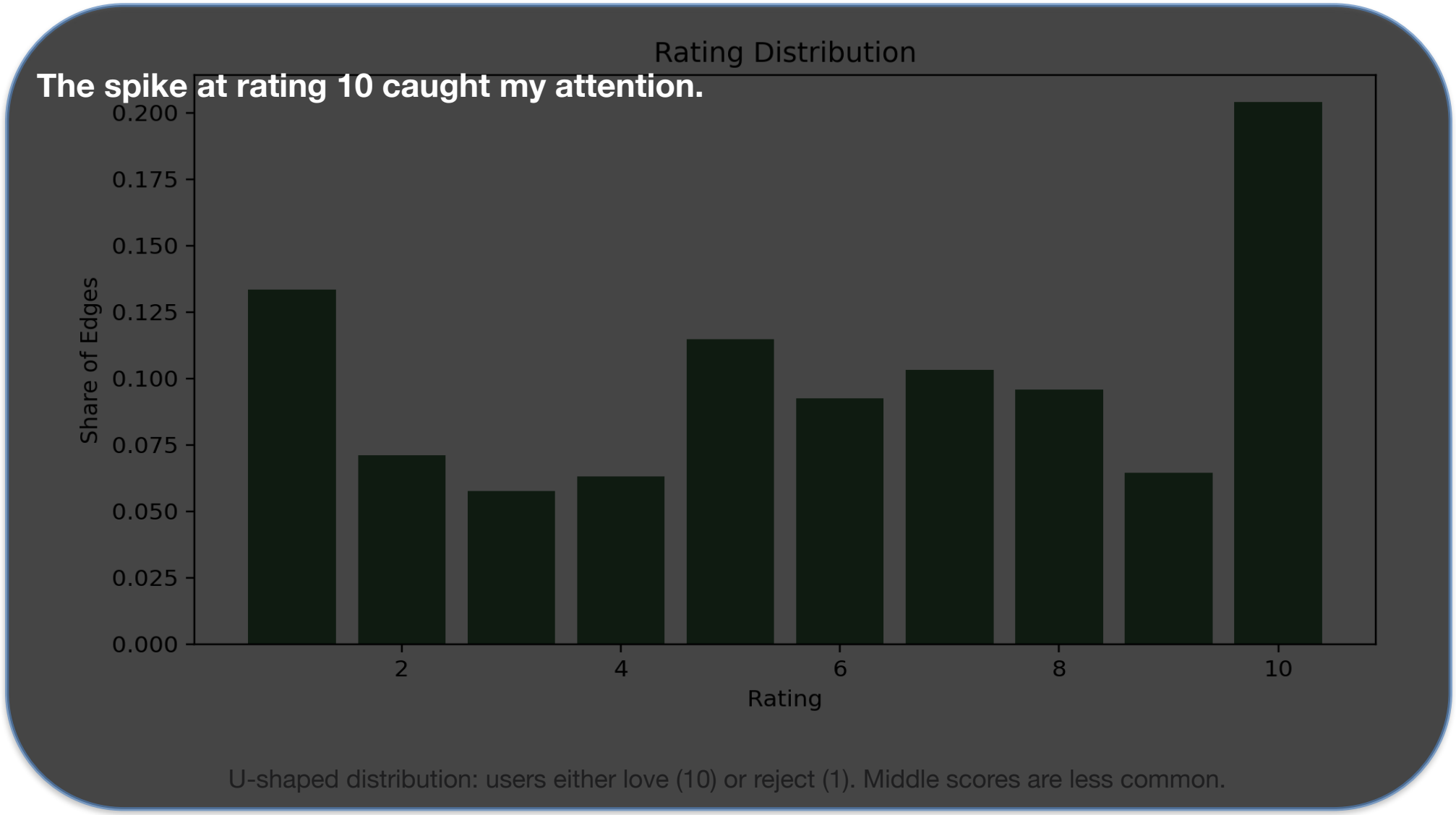
Mean score (range 1–10)

Rating Distribution



U-shaped distribution: users either love (10) or reject (1). Middle scores are less common.

Rating Distribution

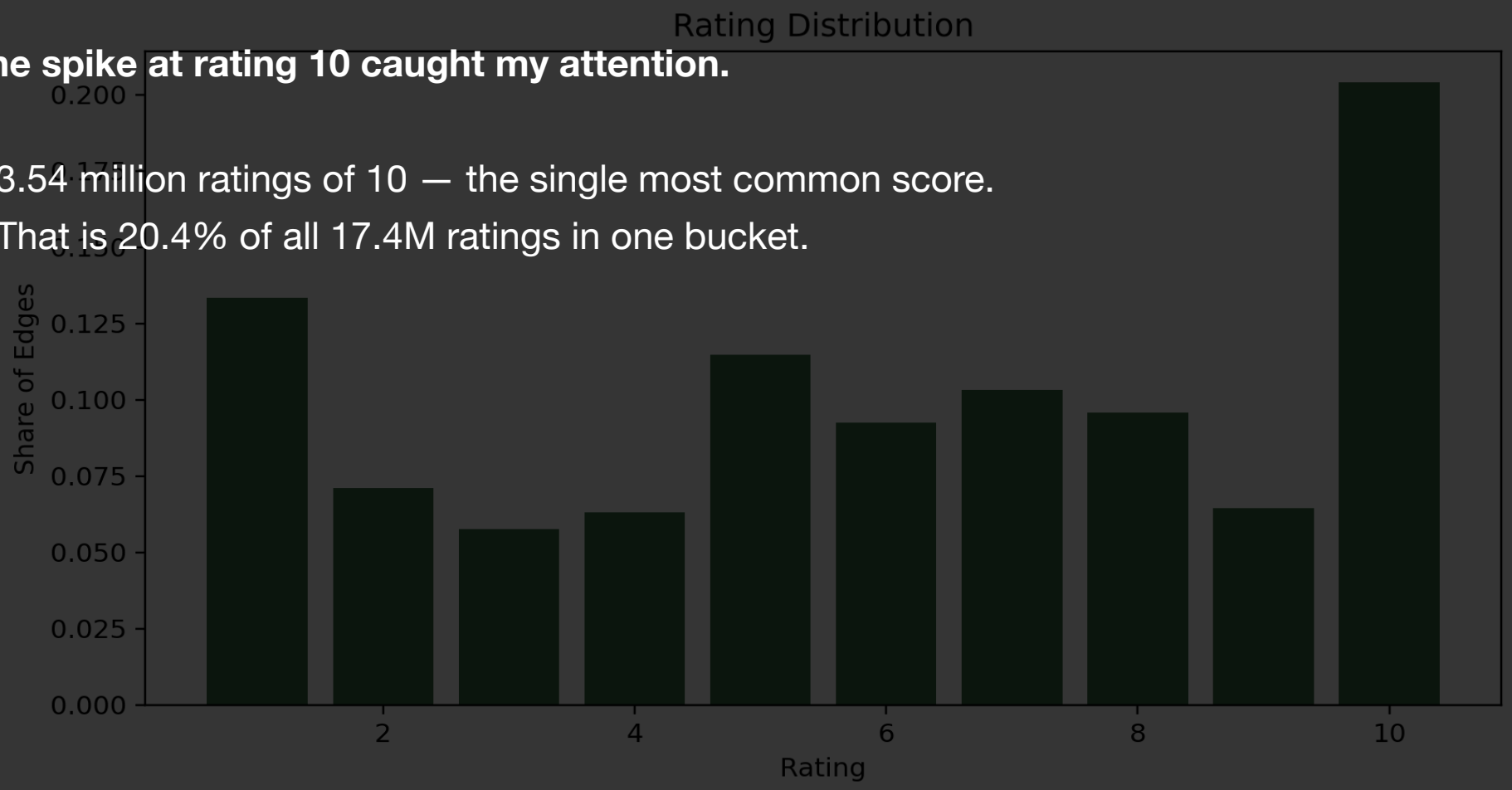


Rating Distribution

The spike at rating 10 caught my attention.

3.54 million ratings of 10 — the single most common score.

That is 20.4% of all 17.4M ratings in one bucket.



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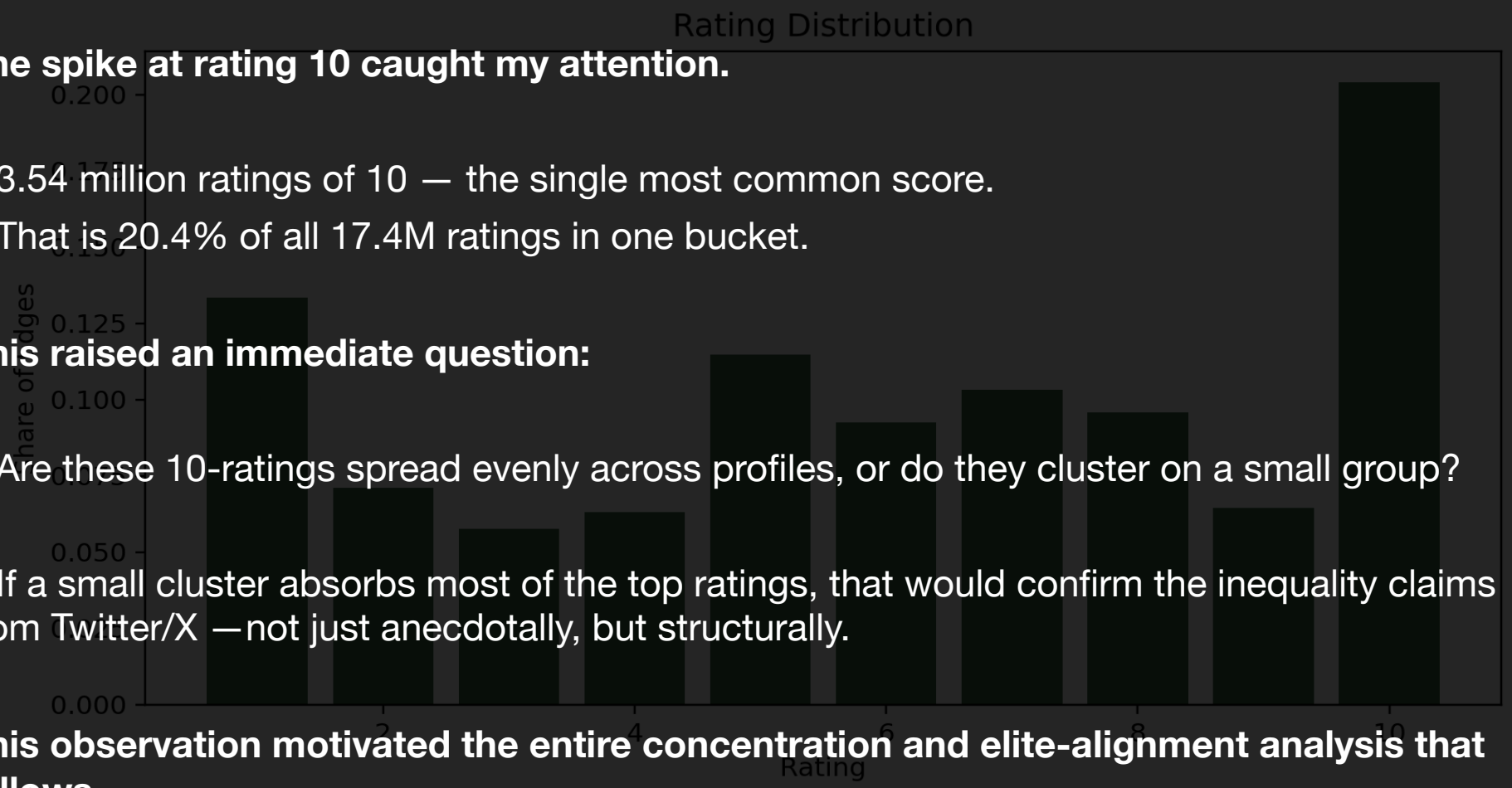

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This raised an immediate question:

Are these 10-ratings spread evenly across profiles, or do they cluster on a small group?

If a small cluster absorbs most of the top ratings, that would confirm the inequality claims from Twitter/X —not just anecdotally, but structurally.

This observation motivated the entire concentration and elite-alignment analysis that follows.

U-shaped distribution: users either love (10) or reject (1). Middle scores are less common.

Feature Engineering: 3 Columns → 7 Features

Raw data — only 3 fields

rater_id → who sends the rating
profile_id → who receives the rating
rating → score from 1 to 10

With so few raw features, I needed to derive meaningful network properties to enable deeper analysis.

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Rater ID	Profile ID	Score
Rater::1	Profile::1	score
Rater::1	Profile::3	score

Feature Engineering: 3 Columns → 7 Features

Why separate roles?

- Same user ID can appear as both rater and profile
- Sending ratings ≠ receiving ratings — structurally different
- Separating avoids false identity assumptions

This lets us apply role-consistent measures like HITS authority on the receiving side.

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Derived features (7 new)

In-degree → visibility
In-strength → volume (breadth + intensity)
Mean rating → evaluative quality
HITS Authority → network prestige
Popularity-rank % → relative standing (volume)
Prestige-rank % → relative standing (prestige)
Prestige gap → prestige vs. popularity mismatch

3 raw columns → 7 meaningful network dimensions

Derived Features — Calculation & Purpose

How each feature is built and what it answers

In-degree = # distinct raters of a profile

→ How many people noticed this profile?

In-strength = Σ ratings received

→ Total volume of attention (breadth × intensity).

Mean rating = in-strength / in-degree

→ Average evaluative quality per rater.

HITS Authority = iterative score on bipartite graph

→ Prestige: am I rated by raters who themselves matter?

Popularity-rank % = percentile of in-strength

→ Where does the profile stand on raw volume?

Prestige-rank % = percentile of authority

→ Where does it stand on network prestige?

Prestige gap = prestige% – popularity%

→ Is the profile over- or under-prestiged for its volume?

Research Questions

Three linked questions:

1. How strongly do popularity and prestige align on the profile side?
2. How concentrated is received attention?
3. Do the profiles dominating total interactions also dominate high-rating buckets?

Motivated directly by the rating-10 spike:
does that mass of top scores land on a small elite?

Popularity vs. Prestige — Question

The question

Do raw popularity (volume of attention) and network prestige (HITS authority) measure the same thing, or does prestige reveal a different hierarchy?

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Do raw popularity (volume of attention) and network prestige (HITS authority) measure the same thing, or does prestige reveal a different hierarchy?

In other words,

is being rated by many people the same as being rated by people who themselves matter in the network?

Popularity vs. Prestige

Strong alignment, but not identical

Metric	Full	POS8 (≥8)
Pearson	844	870
Spearman	855	799
Top-100 shared	75/100	78/100

The same broad hierarchy appears under both measures.

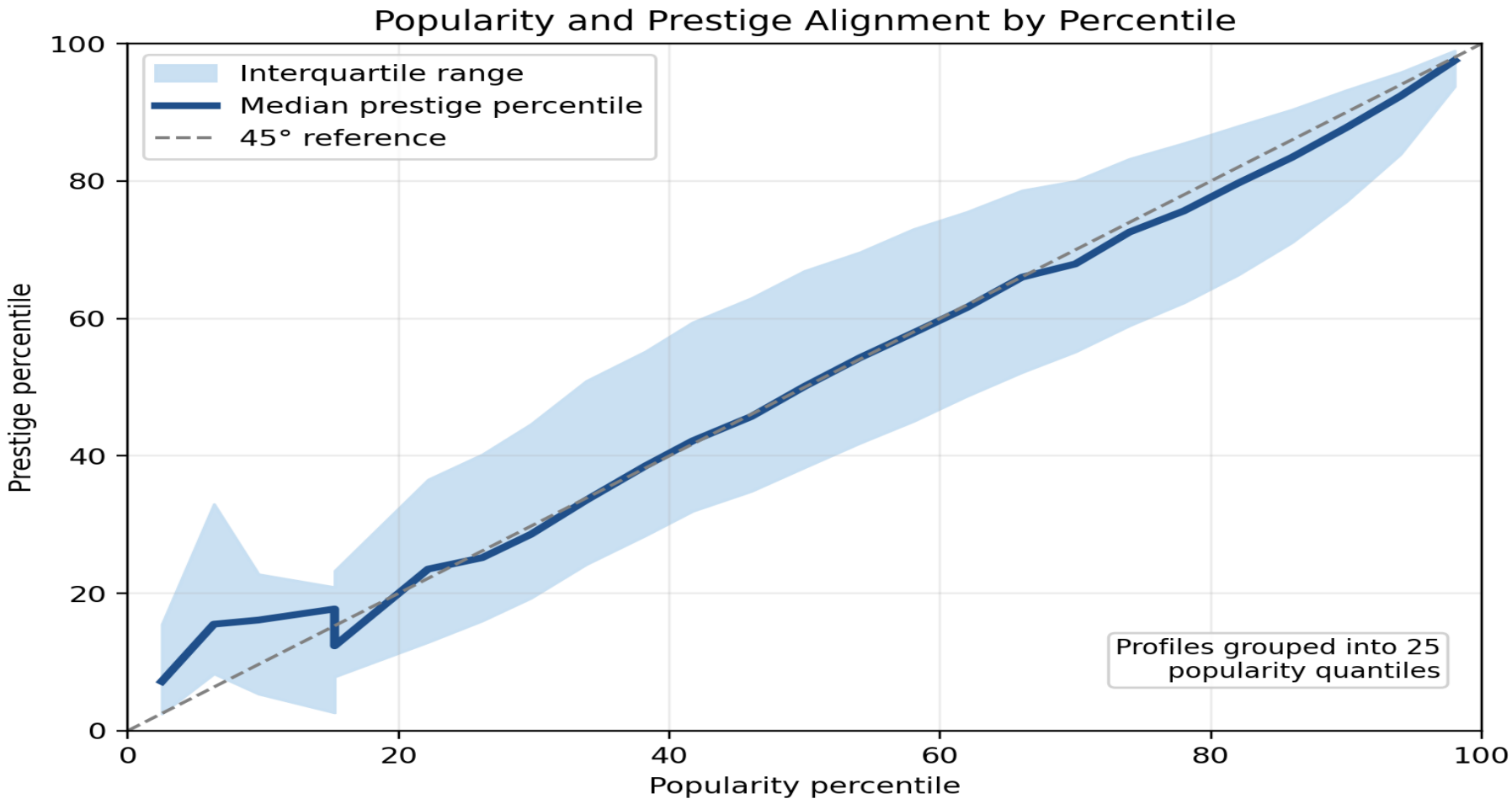
But prestige refines popularity

- 75 of the top-100 are shared, 25 are not
- **Some profiles are popular but not prestigious** (rated by many, but their raters carry little weight in the network)
- **Some profiles are prestigious but not popular** (rated by few, but their raters carry strong network weight)

raw visibility ≠ network position.

Prestige doesn't overturn popularity — it refines it.

Popularity vs. Prestige — Percentile View



Median prestige tracks the 45° line. The interquartile band shows meaningful dispersion remains.

Concentration of Attention

83.9%

of all interactions captured by top 20%

43.7%

of high ratings (≥8) captured by top 1%

0.875

Gini coefficient for high ratings

4.0%

of profiles needed for half of all interactions

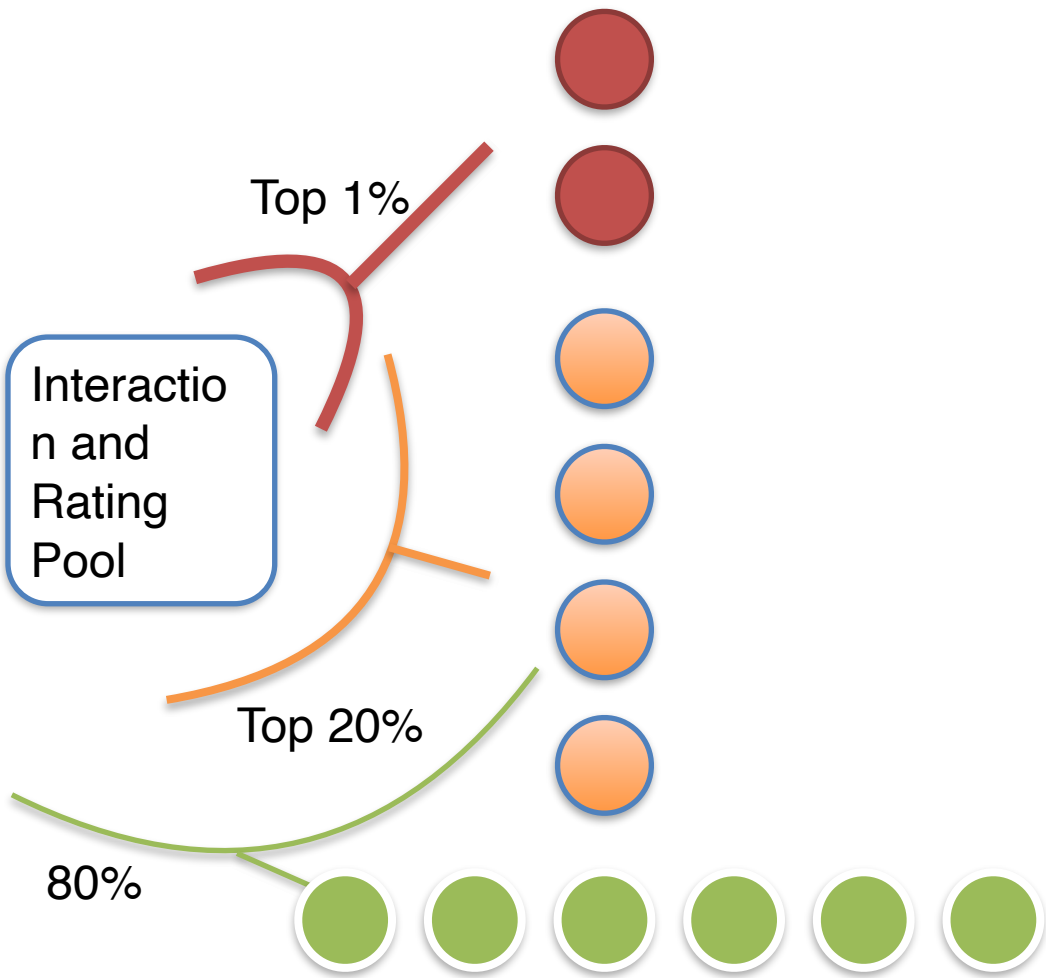
Testing the Claims with Data

Inequality metrics

Gini (interaction count): 0.807
Gini (rating weight): 0.831
Gini (high ratings ≥8): 0.875

Top 1% → 28.2% of all interactions
Top 1% → 43.7% of high ratings
Top 20% → 83.9% of all interactions

Only 1.5% of profiles capture half of all high ratings.



Testing the Claims with Data

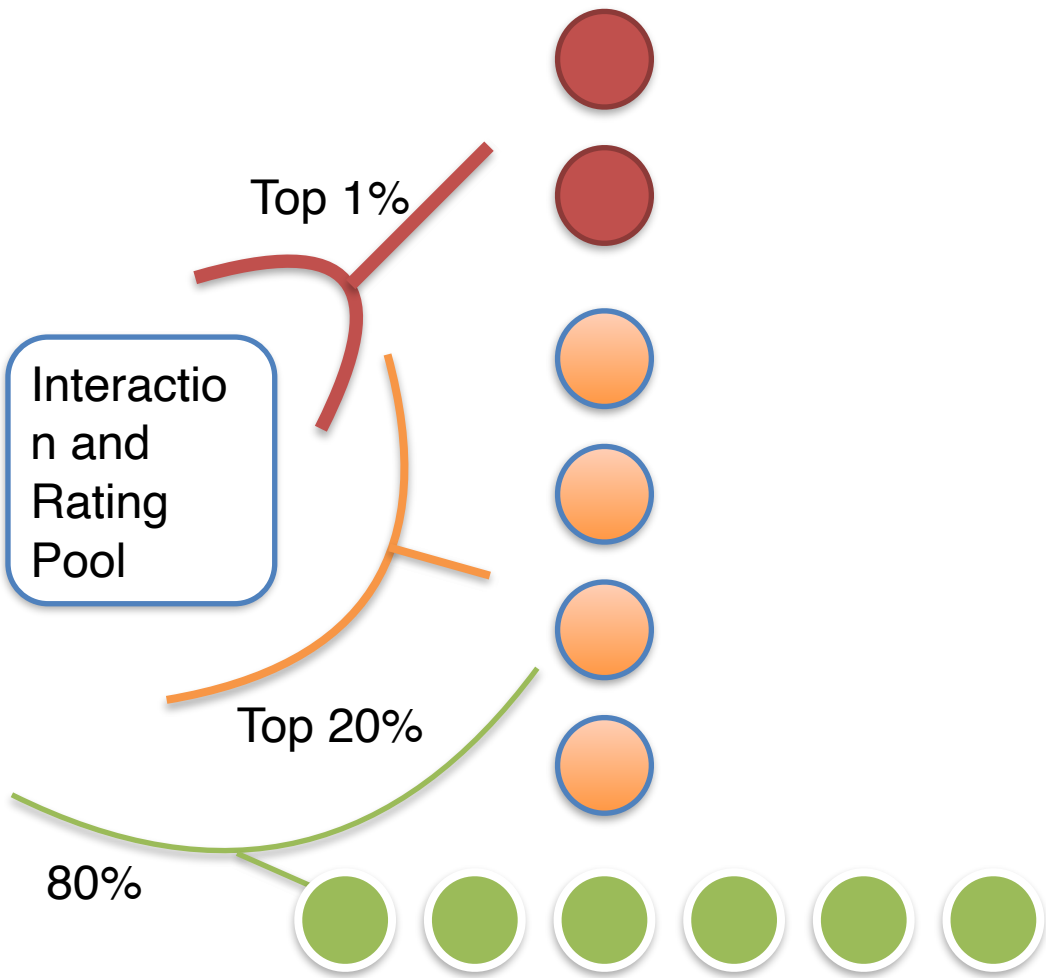
Remember: "Top 20% gets everything"?

Top 20% captures 83.9% — the claim holds.

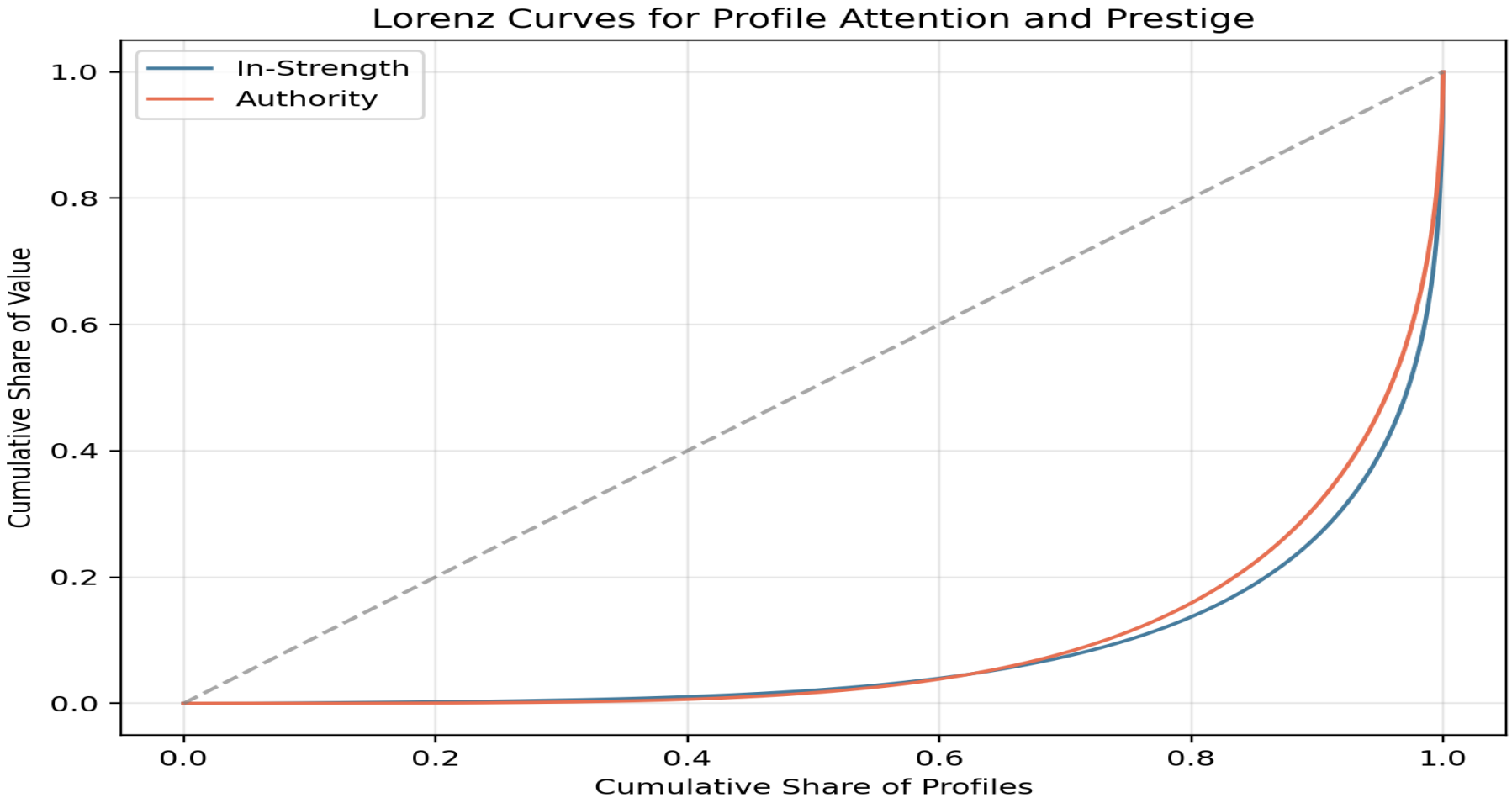
High ratings are even more concentrated than total interactions.

These Gini values exceed most income inequality measures worldwide.

The anecdotal claims from Twitter/X are directionally correct.

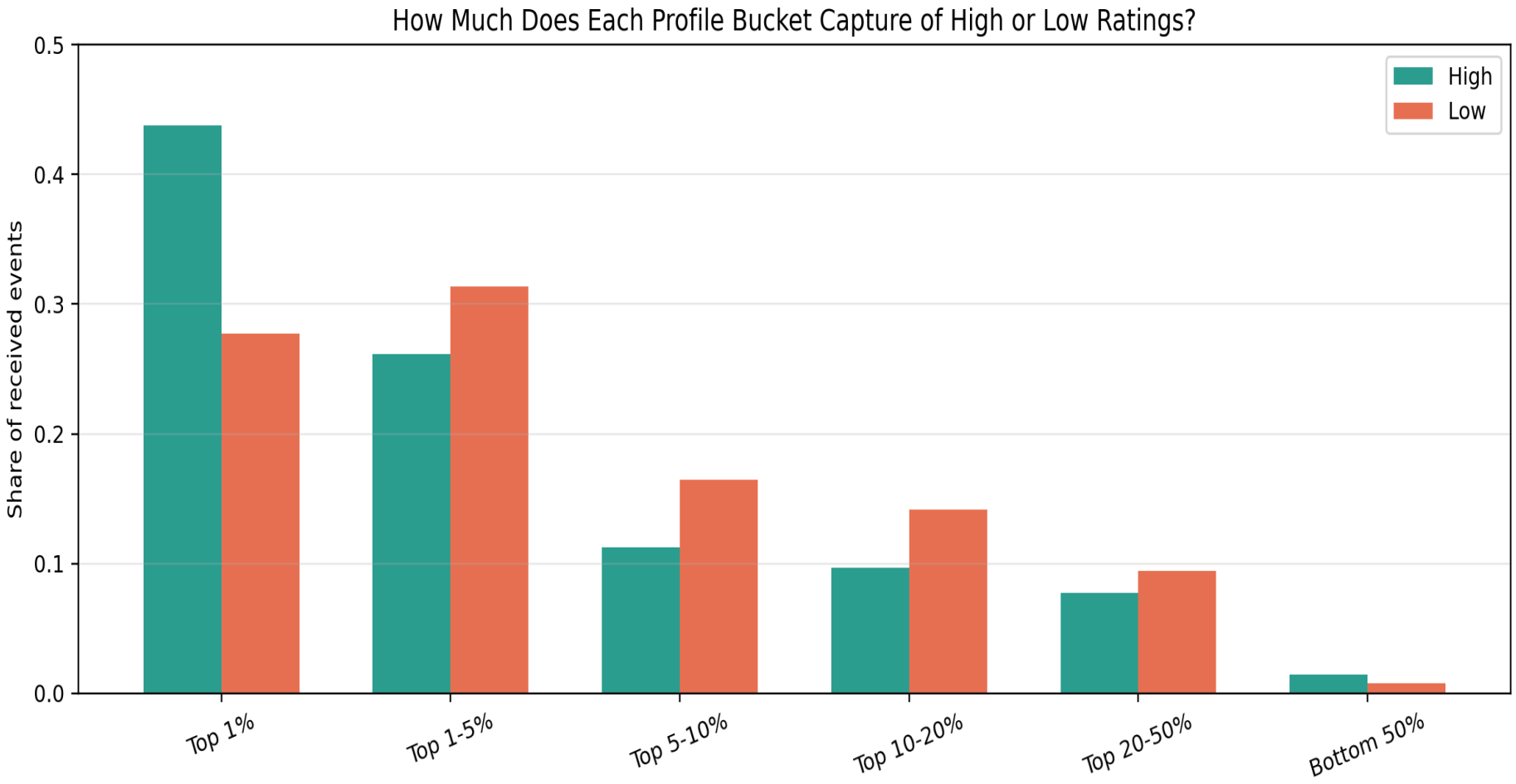


Lorenz Curves — Visualizing Inequality



Both in-strength and authority are far from the equality diagonal. Raw attention is slightly more concentrated.

Profile Bucket Shares



Top 1% absorbs ~28% of all interactions and ~44% of high ratings — attention is captured by a tiny elite.

Degree Distribution Fit

Power-law tail analysis (Clauset et al. 2009)

Fitted exponent: α = 2.51 (σ = 0.02)

Tail starts at: x_min = 613

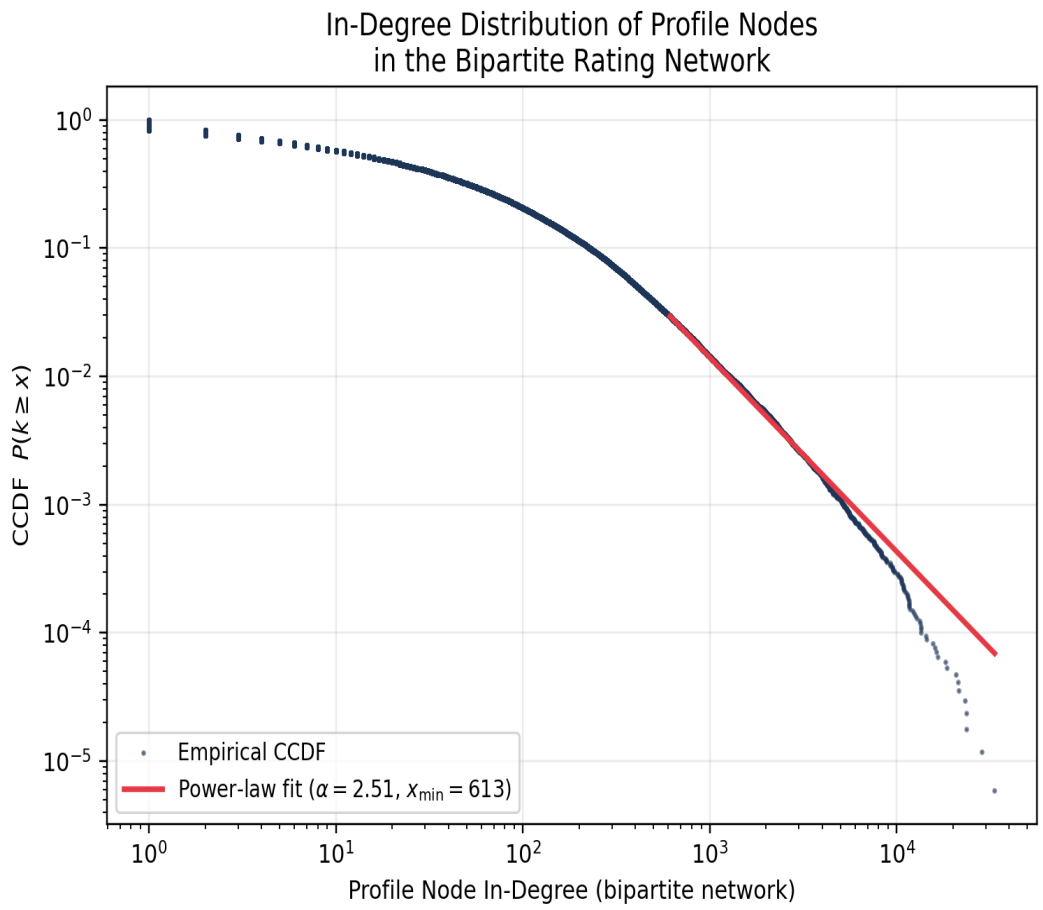
Nodes in tail: 4,971 profiles

Model comparison

- Exponential rejected (R = 868.1, p ≈ 0)
- Log-normal fits better globally (R = −12.4, p = 0.002)

Typical for real networks: power-law upper tail, log-normal body.

The tail is genuinely heavy.



Bucket Shares — Who Gets What?

Profile bucket shares

Bucket	All	High (≥8)	Low (≤3)
Top 1%	28.2%	43.7%	27.7%
Top 1–5%	25.9%	26.1%	31.3%
Top 5–10%	14.8%	11.3%	16.5%
Top 10–20%	15.0%	9.7%	14.2%
Top 20–50%	13.8%	7.8%	9.5%
Bottom 50%	2.2%	1.4%	0.8%

Top 1% captures almost half of all high ratings.

Elite profiles also attract low ratings —consistent with an exposure mechanism.

Elite Alignment — The Main Contribution

Top 1% interaction bucket and top 1% high-rating bucket overlap on:

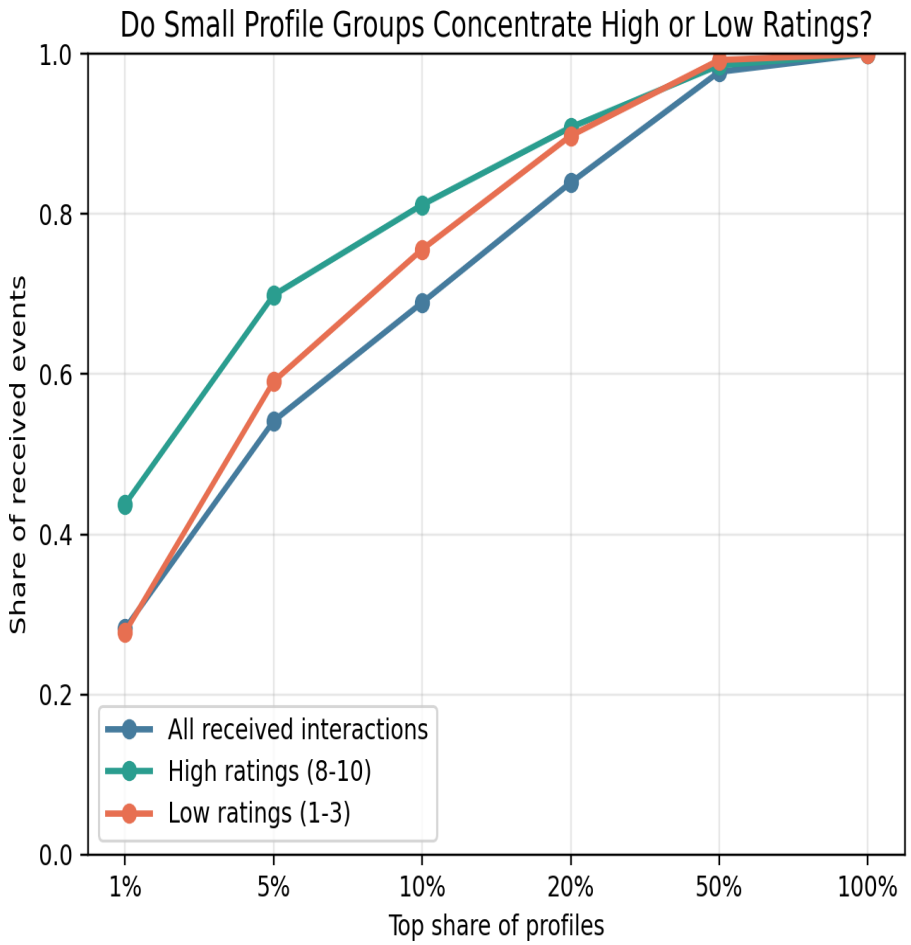
1,029 profiles (out of 1,687)

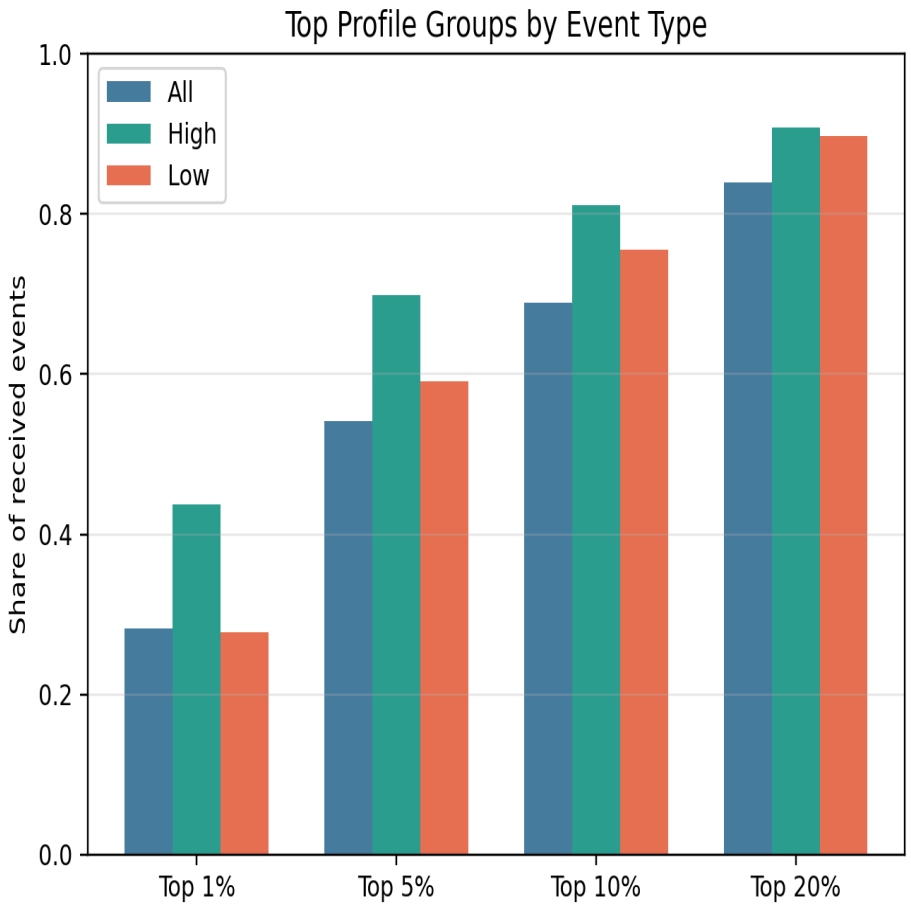
Metric	Value
Shared (%)	61.0%
Jaccard	439
Expected overlap	16.9 profiles
Max (20,000 draws)	37 profiles
Observed	1,029 profiles

$p < 10^{-1700}$

Real overlap is 60× the random expectation.
This is not chance — it is structural.

Elite Alignment — The Main Contribution





1,029 profiles overlap between the top-1% interaction and top-1% high-rating buckets — 60× the random expectation.

Which Features Predict Elite Membership?

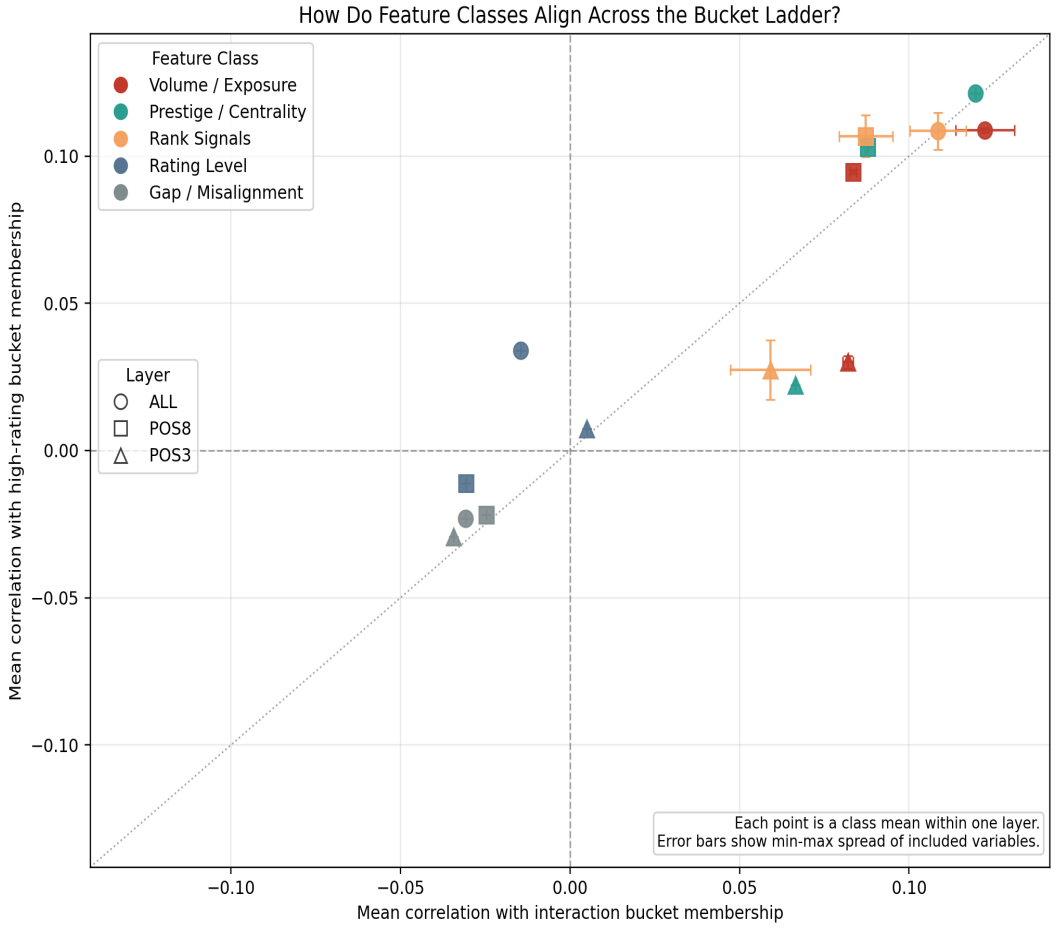
Point-biserial correlations (Top 1%)

Feature	Interaction	High-rating
In-degree	0.68	0.56
In-strength	0.64	0.62
Authority	0.59	0.60
Mean rating	0.03	0.09
Prestige gap	-0.02	-0.01

In-degree: strongest single predictor.

Authority: most balanced across both sides.

Mean rating tells almost nothing — a profile can attract thousands of raters and still have a moderate average. Elite status is about visibility, not average score.



Conclusions

Three key findings

1. Popularity and prestige are strongly aligned but not identical
   → keeping both measures is worthwhile

2. Attention is extremely concentrated
   → top 20% captures 83.9% | Gini = 0.807
   → high ratings even more concentrated | Gini = 0.875

3. Most visible and most positively rated profiles share the same structural core
   → 1,029 vs. 16.9 expected — far beyond chance

The claims about dating inequality are directionally correct —but the real story is more nuanced: it's about network position, not just average rating.

Limitations

- Purely descriptive — no causal claims
- Anonymous data — no demographics (gender, age, etc.)
- Ratings ≠ reciprocal matches (real dating outcomes unknown)
- Only 3 raw columns — richer data would enable deeper analysis

Thank You

Questions?

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